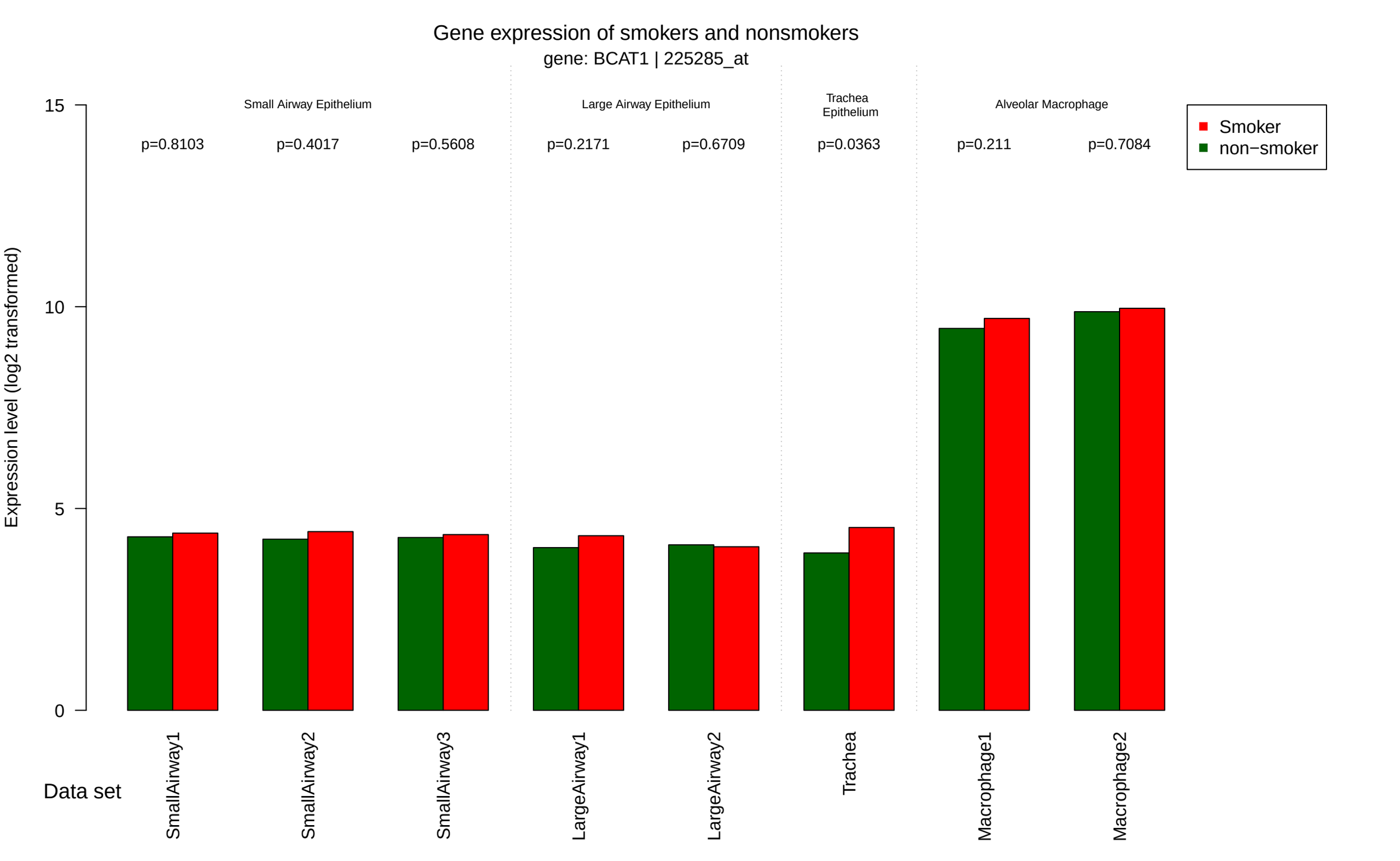
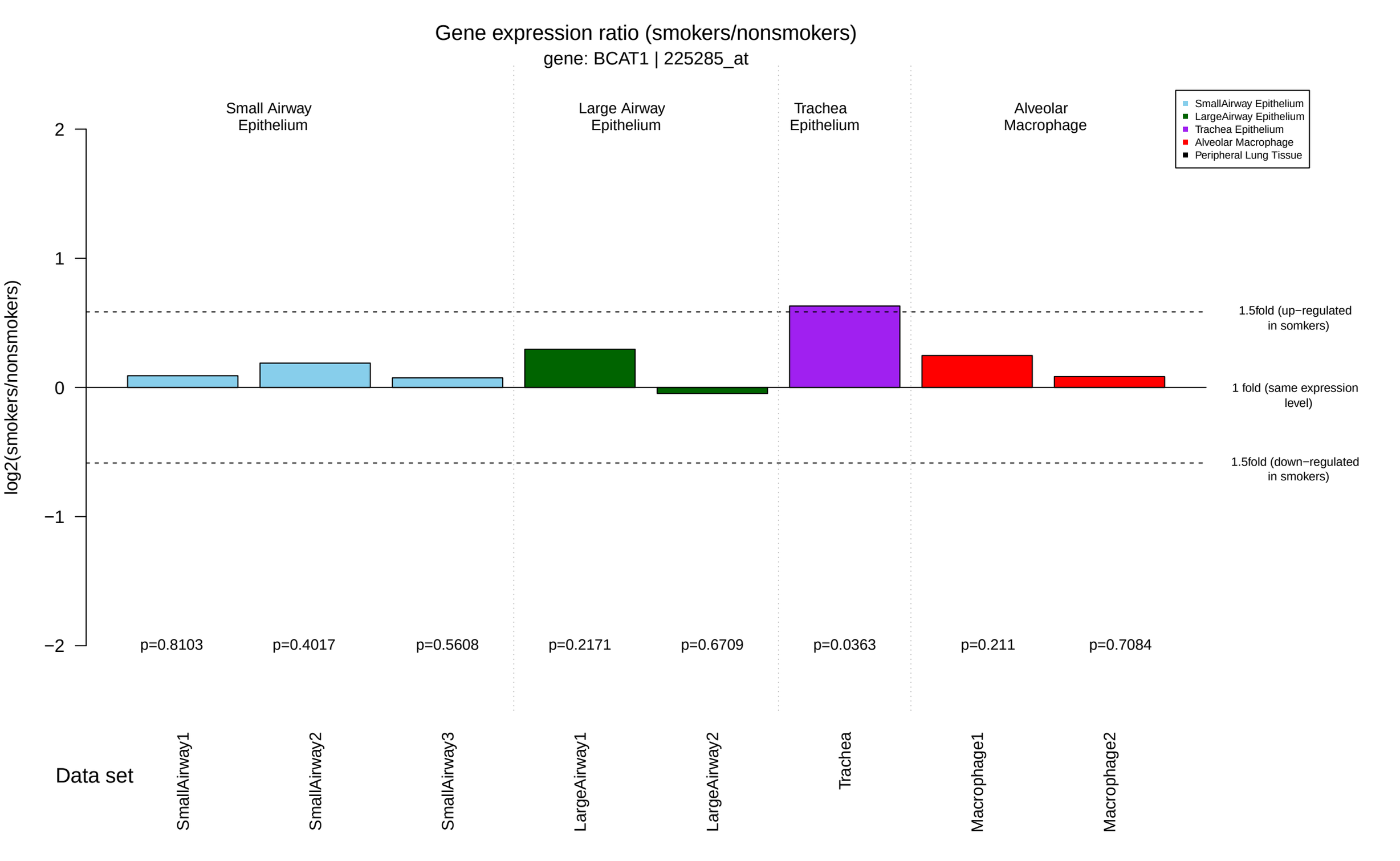
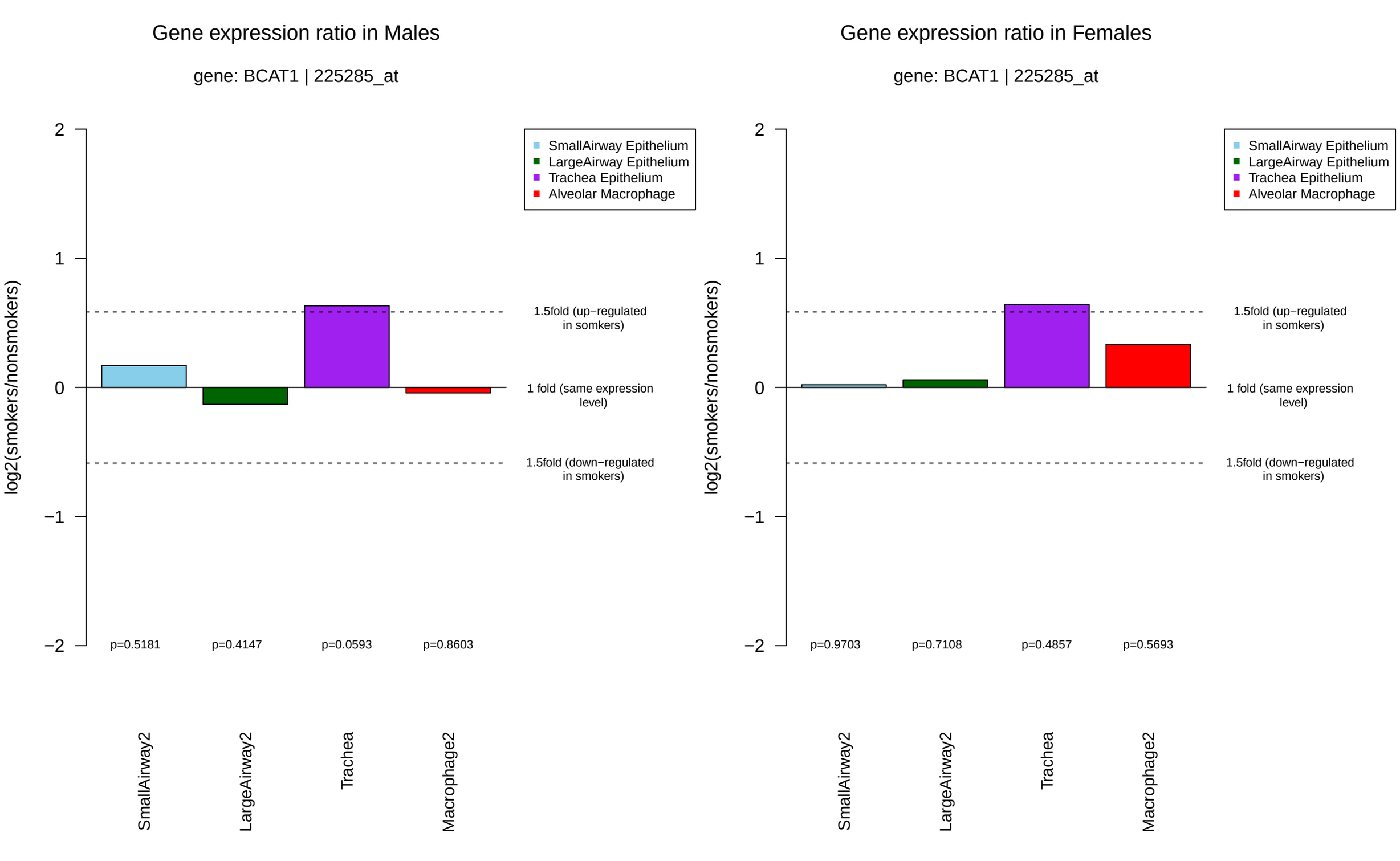
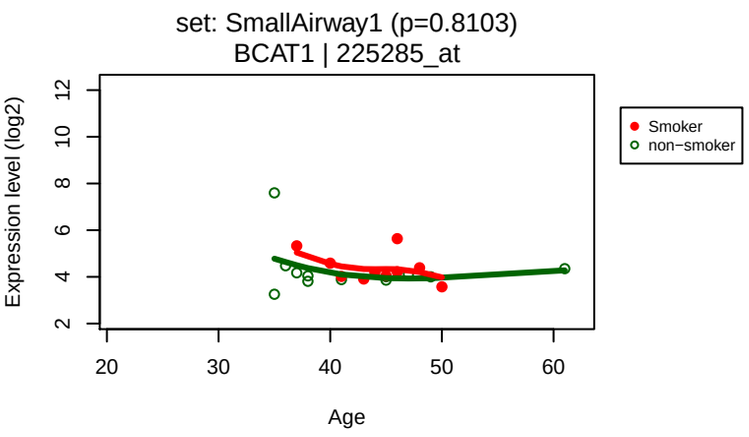
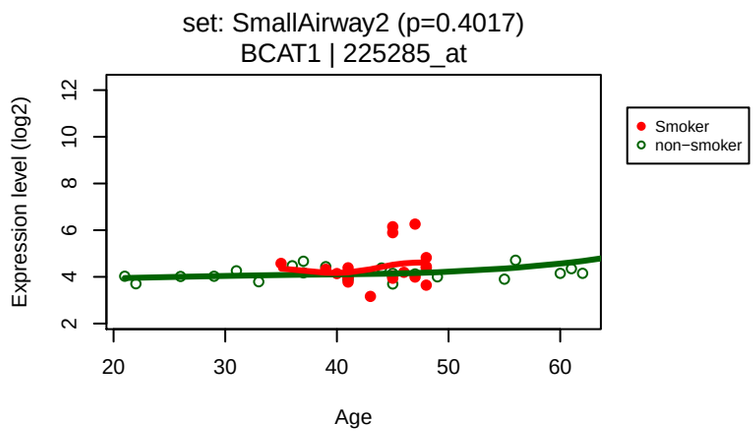
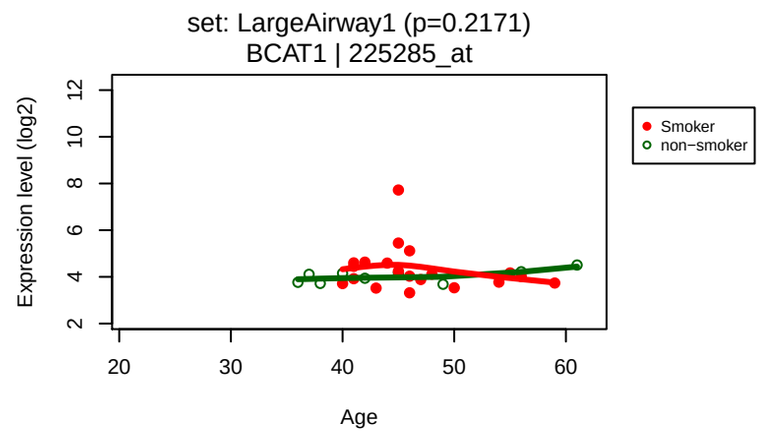

Gene expression of smokers and nonsmokers
gene: BCAT1 | 225285_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8103
p=0.4017
p=0.5608
p=0.2171
p=0.6709
p=0.0363
p=0.211
p=0.7084
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

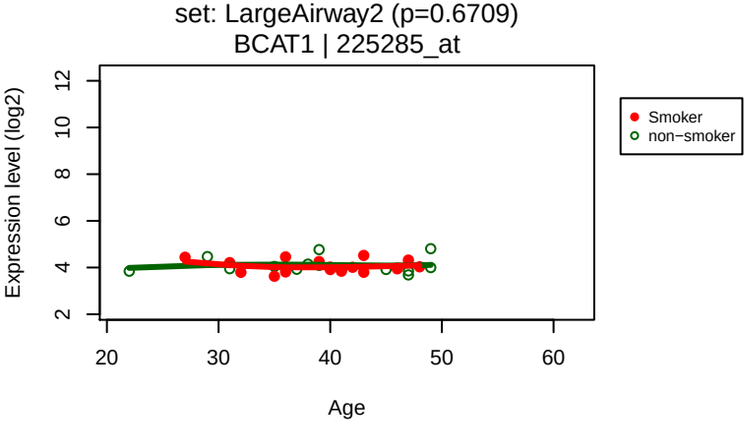

Gene expression ratio (smokers/nonsmokers)
gene: BCAT1 | 225285_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8103
p=0.4017
p=0.5608
p=0.2171
p=0.6709
p=0.0363
p=0.211
p=0.7084
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

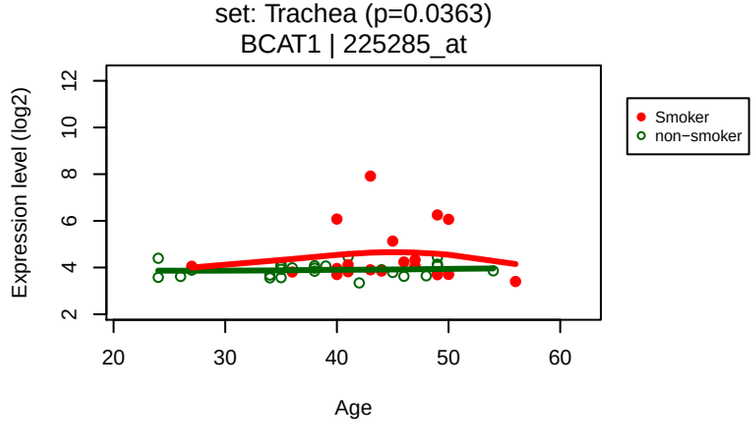

Gene expression ratio in Males
gene: BCAT1 | 225285_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5181
p=0.4147
p=0.0593
p=0.8603
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BCAT1 | 225285_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9703
p=0.7108
p=0.4857
p=0.5693
SmallAirway2
LargeAirway2
Trachea
Macrophage2

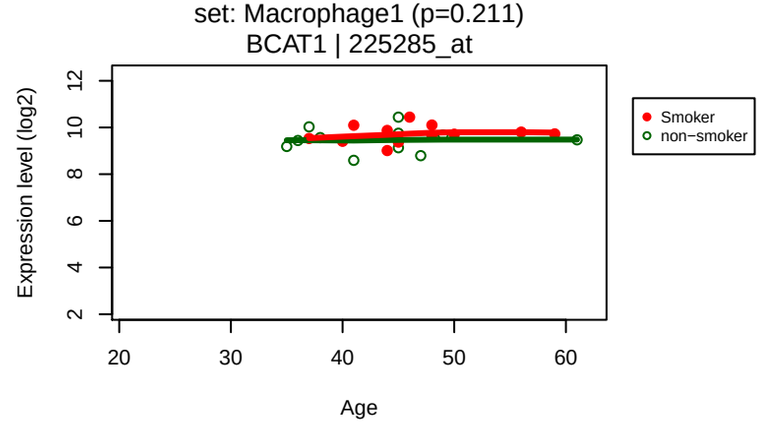

set: SmallAirway1 (p=0.8103)
BCAT1 | 225285_at
Expression level (log2)
Age
Smoker
non−smoker

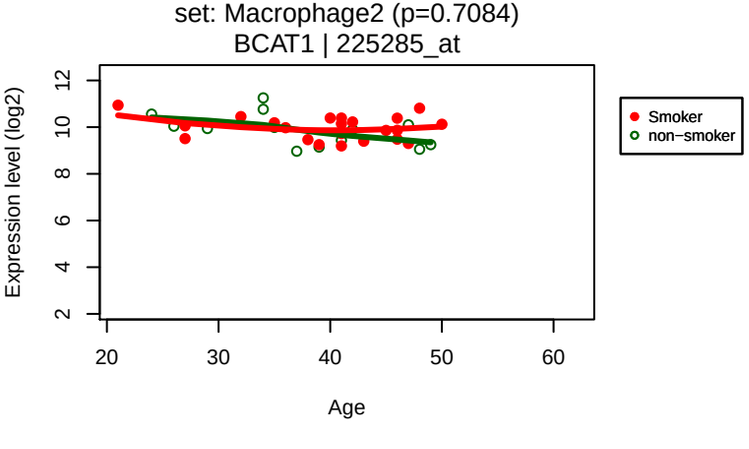

set: SmallAirway2 (p=0.4017)
BCAT1 | 225285_at
Expression level (log2)
Age
Smoker
non−smoker

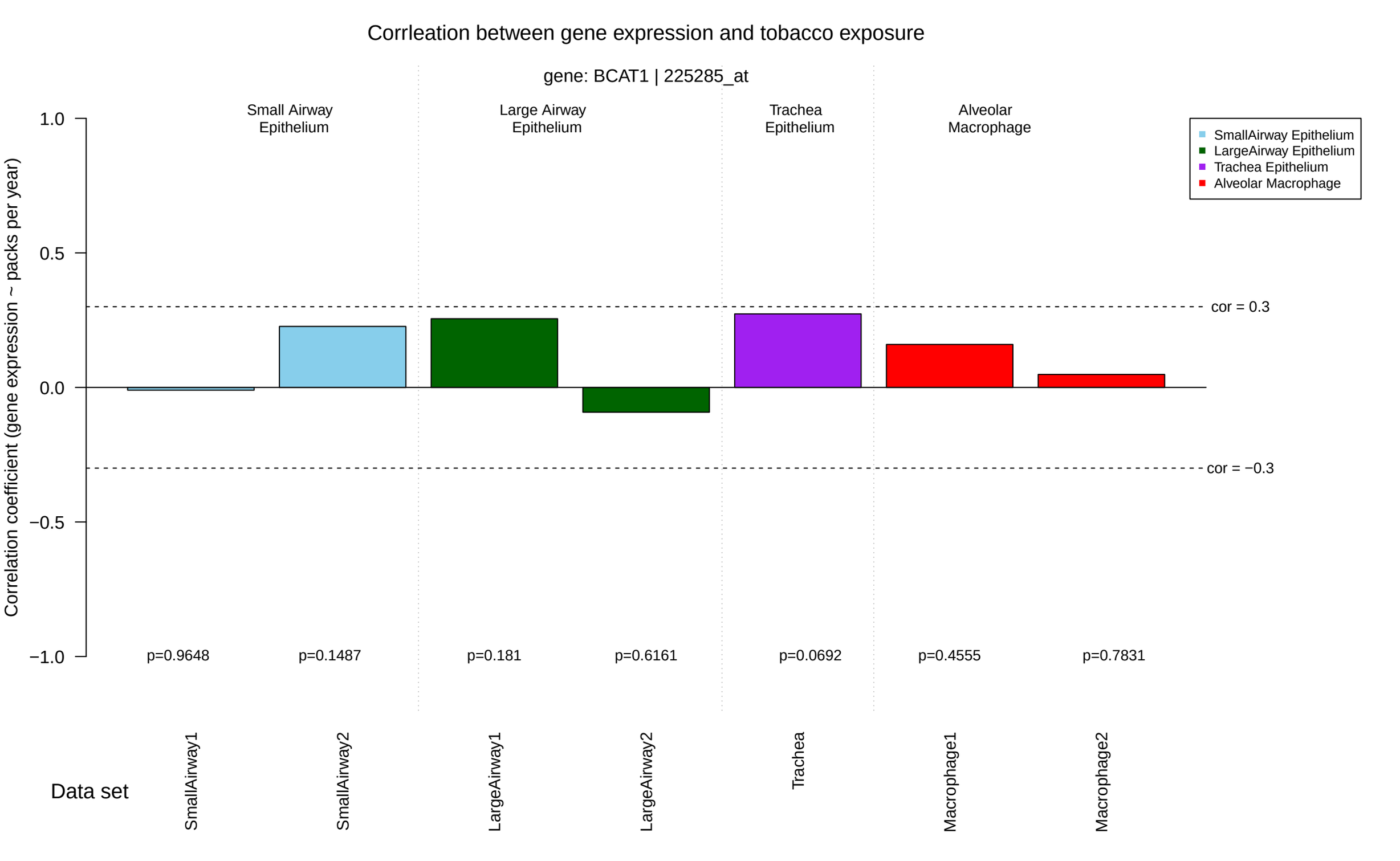

set: LargeAirway1 (p=0.2171)
BCAT1 | 225285_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6709)
BCAT1 | 225285_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0363)
BCAT1 | 225285_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.211)
BCAT1 | 225285_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7084)
BCAT1 | 225285_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BCAT1 | 225285_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9648
p=0.1487
p=0.181
p=0.6161
p=0.0692
p=0.4555
p=0.7831
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2